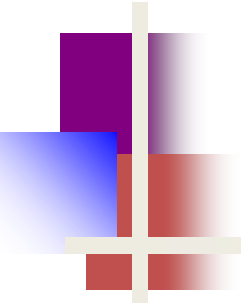


2026 제7회 전자구조계산 여름학교



SIESTA로 배우는 원자 오비탈 기저함수 DFT 코드: **SIESTA 실습**

실습 조교: 이룡규 박사
보조: 강현준, 서영진, 이정원

*School of Electrical Engineering
Korea Advanced Institute of Science & Technology*

2026. 7. 14 (Tue.)

1. Overview of SIESTA SIESTA 개요 및 기본적인 DFT 계산 절차 대해서 알아본다
2. Basic computational setting SIESTA 계산의 입력/출력 형식 및 계산 방법을 알아본다
3. Convergence test k -point 샘플링을 통한 총-에너지 수렴성 테스트 과정을 알아본다
4. Geometrical optimization 구조 최적화 과정을 알아본다
5. Electronic structure analysis 전자구조 계산 및 분석 과정을 알아본다

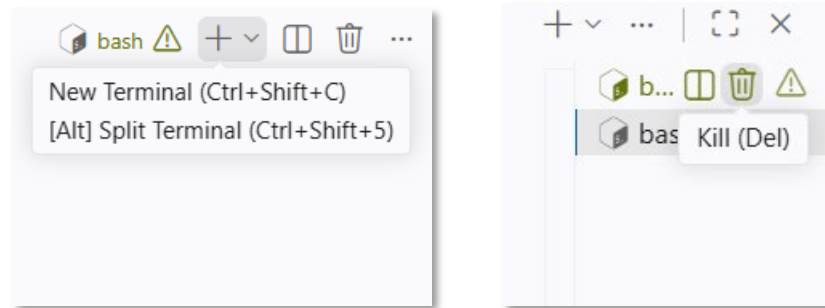
Useful external tutorial: (1) [SIESTA Docs](#)

(2) [YHKLab Wiki](#)

- Enter to [VS Code \(beta\) - NANO - EDISON2](#) (recommended Browser «MS Edge»)
- Initiate Conda environment

```
$ conda init
```

- Open the new session and then kill the old session:



```
(base) XXX@vscode:~$
```

To install the following python script, keep this “base” environment

- Download the “ESC2026 tutorial repository” from GitHub

```
$ git clone https://github.com/YHKlab/ESC2026.git
```

If you already downloaded, just “git pull”

- Change the directory to the “SIESTA tutorial” folder.

```
$ cd ESC2026/exe2-siesta
```

- Let's take a look at the files:

```
$ ls
0.preliminary/  1.basic/  2.convergence_test/
3.geometrical_optimization/  4.electronic_structures/
```

- Install python scripts:

```
$ cd 0.preliminary/
$ bash setup.sh/
$ source ~/.bashrc
$ conda activate conda542
```

Overview of SIESTA

➤ [Wiki Website](#)

SIESTA 개요 및 기본적인 DFT 계산 절차 대해서 알아본다

- SIESTA program¹:



Linear combination of atomic orbital (LCAO) 기저함수 기반 density functional theory (DFT) 계산 코드

- Key features:

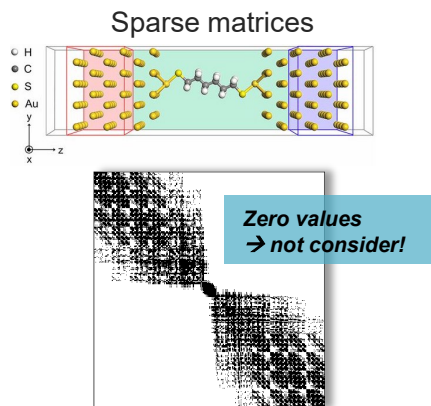
Norm-conserving pseudopotential – *Lecture 1 “Pseudopotential 및 pseudoatomic orbital”*

Strictly confined numerical orbital basis – *Lecture 3 “SIESTA로 배우는 원자 오비탈 기저함수 DFT 코드”*

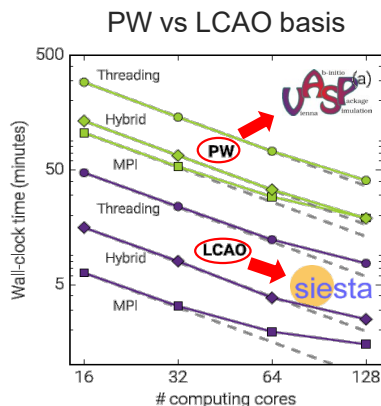
Quantum transport simulation (combined with non-equilibrium Green function (NEGF)) – *TranSIESTA*

- (1) Strictly confined → sparse matrices
- (2) Numerical orbital basis → reduced number of basis

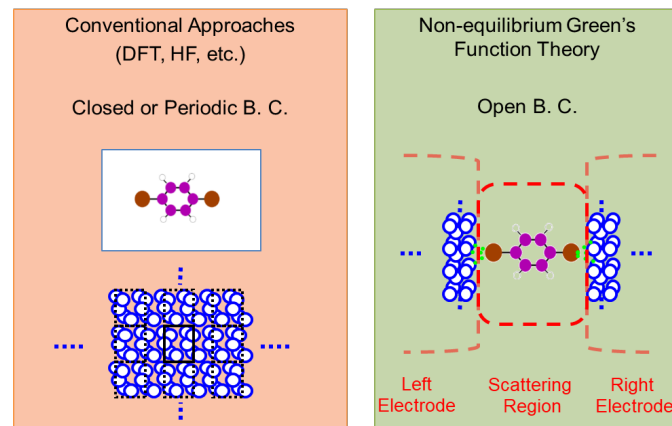
DFT vs DFT-NEGF
→ Equilibrium vs non-equilibrium



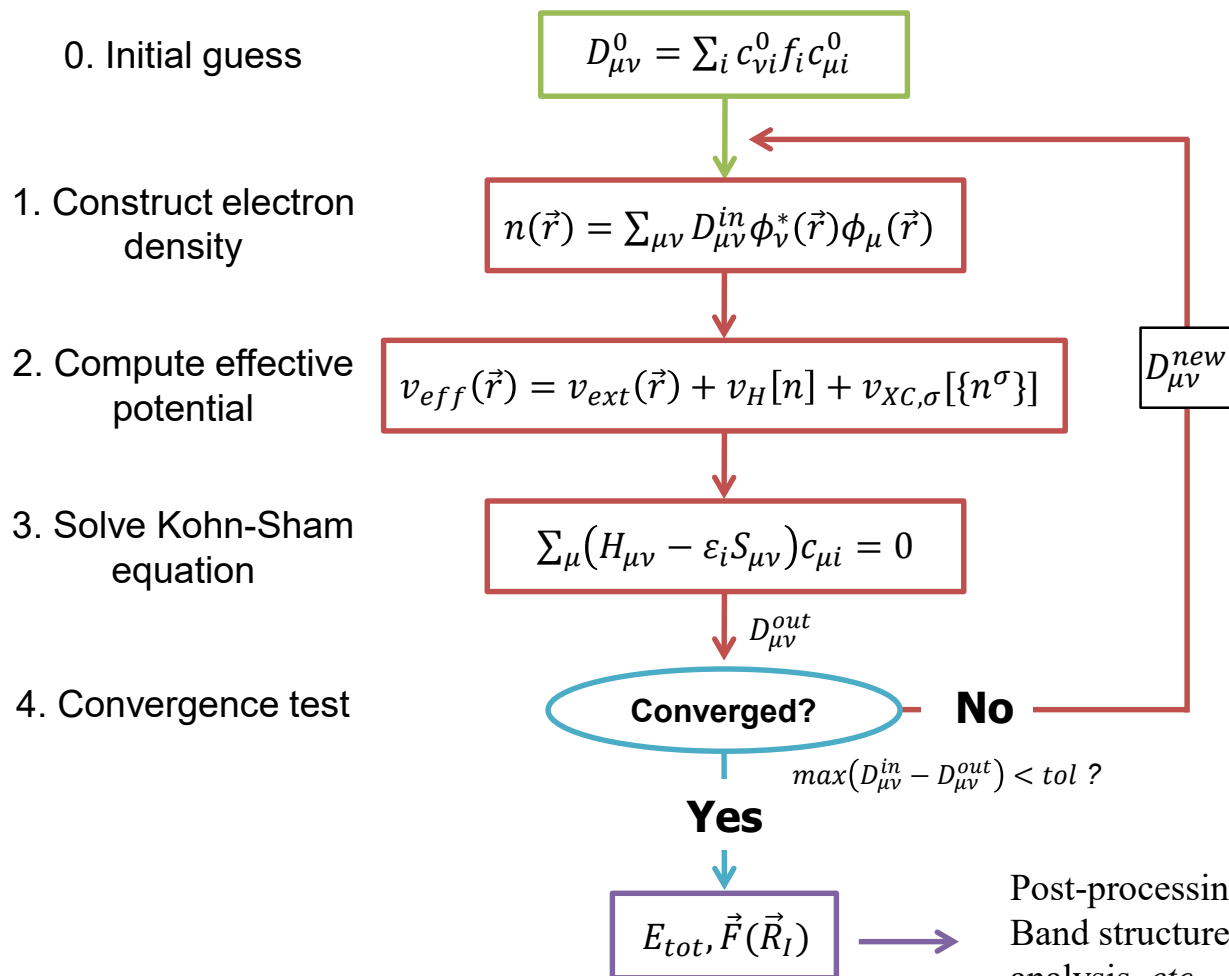
Density, Hamiltonian, Overlap matrices



J. Phys.: Condens. Matter., **32**, 015901 (2020)



Self-consistent field (SCF) process



LCAO formalism

Wavefunction:

$$\psi_i(\vec{r}) = \sum_\mu c_{\mu i} \phi_\mu(\vec{r})$$

Density matrix:

$$D_{\mu\nu} = \sum_i c_{\nu i} f_i c_{\mu i}$$

f_i : Occupation of ψ_i



- **Modeling of (initial) atomic structure:**
전자구조 특성 계산의 대상인 원자 구조(원소, 위치)를 모델링하는 과정
- **Basis-set optimization (*optional*):**
DFT 계산의 LCAO 기저함수의 최적화 과정;
- **Brillouin zone (BZ) (or *k*-points) sampling :**
주기적 경계조건에 따른 Brillouin zone (BZ) 에 대한 수치적인 *k*-points 샘플링 과정
- **Geometrical optimization:**
총 에너지, 원자힘 계산을 통한 안정한 구조를 찾는 과정; 격자 상수(lattice constant) 및 원자 위치
- **Electronic structure analysis:**
최종적으로 얻은 계산 조건 및 구조를 통해 밴드구조, 상태밀도함수 등 전자구조 계산 및 분석 과정

Basic computational setting

➤ [Wiki Website](#)

SIESTA 계산의 입력/출력값 형식 및 계산 방법을 알아본다

- Change the directory to the “1.basic/H2O/input” folder

```
$ cd 1.basic/H2O/input
$ ls
RUN.fdf  STRUCT.fdf  BASIS.fdf  KPT.fdf  H.psf  O.psf
```

“Main” input file

- Exchange-correlation (XC)
- Self-consistent field (SCF)
- Grid sampling
- I/O options
- Etc.

“Sub” input files (*optional**)

- Structural information
- Basis information
- *k*-point sampling information

Pseudopotential files (psf, ion, psml)

- Pseudopotential information per each chemical species

How to obtain?

- 1) Generate by user (ATOM)
- 2) From DB ([pseudo-dojo](#))

* “Sub” 입력 파일들의 구성은 convention일뿐 RUN.fdf 내에 모든 정보를 포함 가능 (pseudopotential 파일 제외)

1. Calculation info.

2. Structure info.

3. Basis info.

4. k-point info.

1.basic/H2O/RUN.fdf

```
#-----
# FDF for water
#-----
#(1) General system descriptors
SystemName      H2O      System name /label
SystemLabel     H2O      (결과 파일 이름 결정)

%include STRUCT.fdf
%include KPT.fdf      Sub input 파일 정보 (optional)
%include BASIS.fdf

#(2) DFT, Grid, SCF
XC.functional    LDA      교환-상관 에너지범함수
XC.authors       CA

MeshCutoff       500.0 Ry  실공간 그리드(grid)
                               샘플링 간격

MaxSCFIterations 500
SCF.Mixer.Method Pulay
DM.MixingWeight  0.05
DM.NumberPulay   10
SCF.Converge.DM   true
SCF.Tolerance.DM  1.d-4
SCF.Converge.H    true
SCF.Tolerance.H   1.d-3 eV

SCF 관련 옵션
(mixer, 수렴 조건 등)
```

1. Calculation info.

2. Structure info.

3. Basis info.

4. k-point info.

1.basic/H2O/RUN.fdf

```
#-----
# FDF for water
#-----
#(1) General system descriptors
SystemName      H2O
SystemLabel     H2O

%include STRUCT.fdf
%include KPT.fdf
%include BASIS.fdf

#(2) DFT, Grid, SCF
XC.functional    LDA
XC.authors       CA
MeshCutoff      500.0 Ry

MaxSCFIterations 500
SCF.Mixer.Method Pulay
DM.MixingWeight  0.05
DM.NumberPulay   10
SCF.Converge.DM   true
SCF.Tolerance.DM 1.d-4
SCF.Converge.H    true
SCF.Tolerance.H   1.d-3 eV
```

교환-상관 에너지범함수

- Available XC functional in SIESTA:

$$\hat{h}_{KS}^0 \psi_{i\sigma}^{\vec{k}} = \left[-\frac{1}{2} \nabla^2 + \hat{v}_{ext} + \hat{v}_H + \hat{v}_{xc,\sigma} \right] \psi_{i\sigma}^{\vec{k}}$$

XC.functional	LDA	GGA
XC.authors	CA, PZ	PBE, PBEsol, BLYP, Etc

1. Calculation info.

2. Structure info.

3. Basis info.

4. k-point info.

1.basic/H2O/RUN.fdf

```
#-----
# FDF for water
#-----
#(1) General system descriptors
SystemName          H2O
SystemLabel          H2O

%include STRUCT.fdf
%include KPT.fdf
%include BASIS.fdf

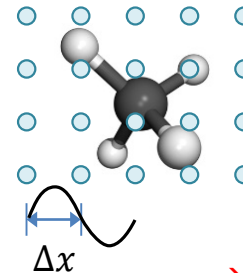
#(2) DFT, Grid, SCF
XC.functional        LDA
XC.authors            CA

MeshCutoff            500.0 Ry

MaxSCFIterations      500
SCF.Mixer.Method      Pulay
DM.MixingWeight        0.05
DM.NumberPulay         10
SCF.Converge.DM        true
SCF.Tolerance.DM       1.d-4
SCF.Converge.H         true
SCF.Tolerance.H        1.d-3 eV
```

실공간 그리드(grid)
샘플링 간격

- Maximum kinetic energy (E_c) of the plane waves that can be represented in the grid without aliasing



$$E_c = \frac{\hbar^2 k_c^2}{2m_e}, \text{ where } k_c = \frac{\pi}{\Delta x}$$

→ the larger E_c , the finer Δx (more accurate)

1. Calculation info.

2. Structure info.

3. Basis info.

4. k -point info.

1.basic/H2O/RUN.fdf

```
#-----
# FDF for water
#-----
#(1) General system descriptors
SystemName          H2O
SystemLabel          H2O

%include STRUCT.fdf
%include KPT.fdf
%include BASIS.fdf

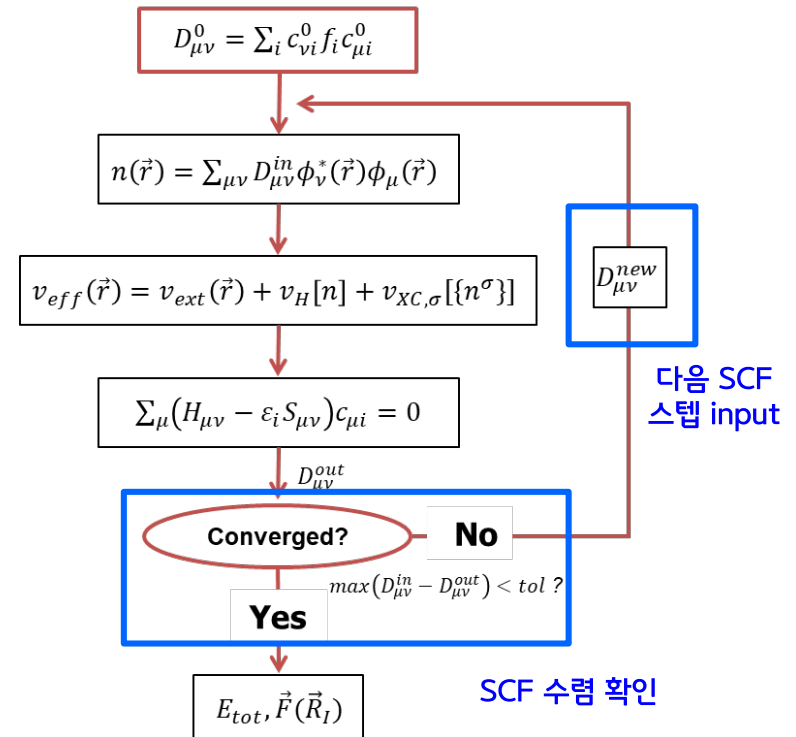
#(2) DFT, Grid, SCF
XC.functional        LDA
XC.authors            CA

MeshCutoff            500.0 Ry
```

```
MaxSCFIterations      500
SCF.Mixer.Method       Pulay
DM.MixingWeight        0.05
DM.NumberPulay         10
SCF.Converge.DM         true
SCF.Tolerance.DM        1.d-4
SCF.Converge.H          true
SCF.Tolerance.H         1.d-3 eV
```

SCF 관련 옵션
(mixer, 수렴 조건 등)

• SCF process:



1. Calculation info.

2. Structure info.

3. Basis info.

4. k -point info.

1.basic/H2O/STRUCT.fdf

```
NumberOfAtoms 3 # Number of atoms
NumberOfSpecies 2 # Number of species
```

```
%block ChemicalSpeciesLabel
1 8 O
2 1 H
%endblock ChemicalSpeciesLabel
```

원소 ID, 원자 번호, 기호*
(기호와 일치하는 psf 파일 必)

```
#(3) Lattice, coordinates, k-sampling
```

단위격자 정보

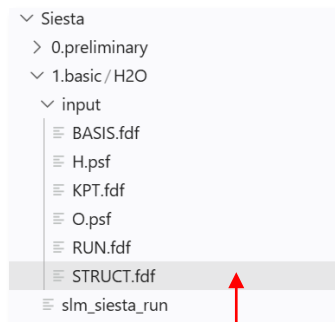
```
LatticeConstant 1.000000000 Ang
%block LatticeVectors
20.000000000 0.000000000 0.000000000
0.000000000 20.000000000 0.000000000
0.000000000 0.000000000 20.000000000
%endblock LatticeVectors
```

```
AtomicCoordinatesFormat Ang
%block AtomicCoordinatesAndAtomicSpecies
10.000000 10.000000 10.298154 1
10.000000 10.763239 10.001845 2
10.000000 9.236761 10.001845 2
%endblock AtomicCoordinatesAndAtomicSpecies
```

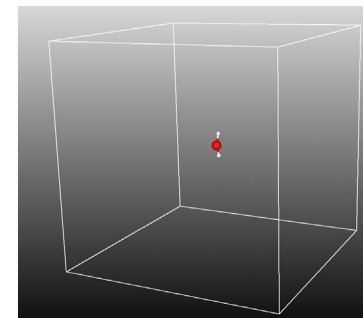
Basis atom의 위치(x,y,z), 원소 ID정보(1 = O, 2 = H)

- Visualize atomic geometry:

Vscode explorer



Crystal Viewer



[Right Click] / [Open in Crystal Viewer]

1. Calculation info.

2. Structure info.

3. Basis info.

4. k -point info.

1.basic/H2O/BASIS.fdf

(1) Basis definition

```
PAO.BasisType      split
PAO.BasisSize      DZP
PAO.EnergyShift     100 meV      # default: 0.02 Ry
PAO.SplitNorm       0.15         # default: 0.15
```

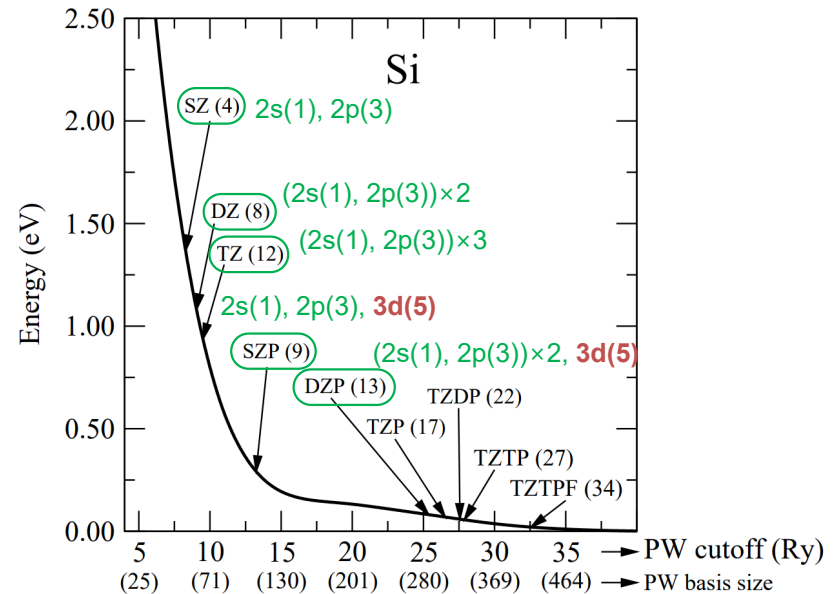
1.basic/H2O/KPT.fdf

```
%block kgrid_Monkhorst_Pack
  1  0  0  0.0
  0  1  0  0.0
  0  0  1  0.0
%endblock kgrid_Monkhorst_Pack
```

- Basis size (number of orbital basis):

PAO.BasisSize SZ, DZ, TZ, SZP, **DZP**, TZP, ...

Optimal Choice



Higher cost, Higher Accuracy

1. Calculation info.

2. Structure info.

3. Basis info.

4. k -point info.

1.basic/H2O/BASIS.fdf

(1) Basis definition

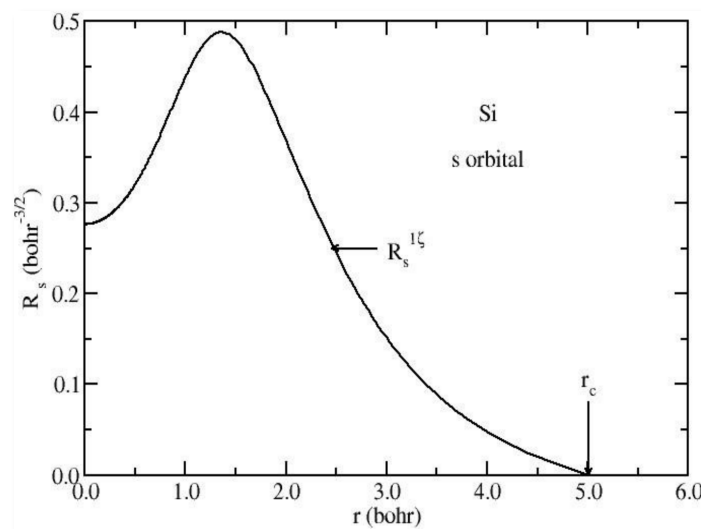
```
PAO.BasisType      split
PAO.BasisSize      DZP
PAO.EnergyShift    100 meV      # default: 0.02 Ry
PAO.SplitNorm      0.15        # default: 0.15
```

1.basic/H2O/KPT.fdf

```
%block kgrid_Monkhorst_Pack
  1  0  0  0.0
  0  1  0  0.0
  0  0  1  0.0
%endblock kgrid_Monkhorst_Pack
```

- Basis range (spatial extension of orbital basis)

The larger energy shift, the longer basis



1. Calculation info.

2. Structure info.

3. Basis info.

4. k -point info.

1.basic/H2O/BASIS.fdf

(1) Basis definition

```
PAO.BasisType      split
PAO.BasisSize      DZP
PAO.EnergyShift    100 meV      # default: 0.02 Ry
PAO.SplitNorm      0.15         # default: 0.15
```

1.basic/H2O/KPT.fdf

```
%block kgrid_Monkhorst_Pack
  1  0  0  0.0
  0  1  0  0.0
  0  0  1  0.0
%endblock kgrid_Monkhorst_Pack
```

분자; 비주기적 → Γ point calculations

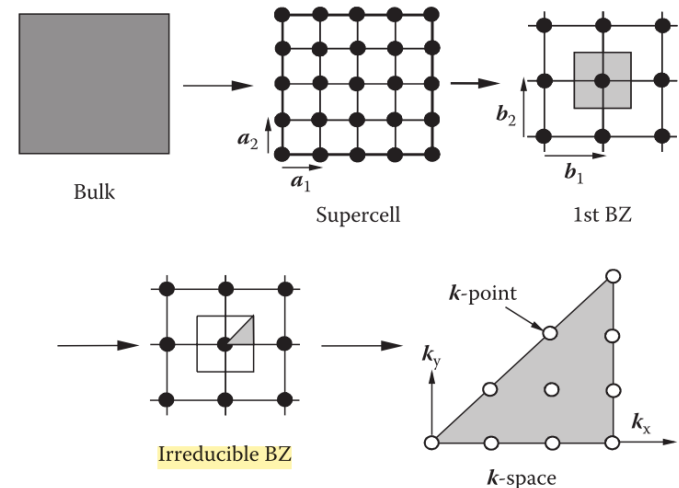
```
%block kgrid.MonkhorstPack
Mk(1,1) Mk(2,1) Mk(3,1) dk(1)
Mk(1,2) Mk(2,2) Mk(3,2) dk(2)
Mk(1,3) Mk(2,3) Mk(3,3) dk(3)
%endblock
```

Reciprocal vector (b_1, b_2, b_3)에
대한 sampling 수

Sampling position shift
(e.g. hexagonal cell,
rectangular cell)

- Brillouin zone sampling with k -points

$$I = \frac{1}{\Omega_{BZ}} \int_{BZ} f(k) d^3k = \sum_{k_i} w_{k_i} f(k_i)$$



<June Gunn Lee, Computational Materials Science : An Introduction>

- Change the directory to the “1.basic/H2O” folder

```
$ cd ..  
$ ls  
input/  slm_siesta_run
```

- Job submission:

```
$ sbatch slm_siesta_run  
$ squeue
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST (REASON)
1125	basic	H2O	rglee	R	0:04	1	slurmd-0

- After calculation done:

```
$ ls  
H2O.1125.err  H2O.1125.out  OUT  input  slm_siesta_run
```

- Change the directory to the “1.basic/H2O/OUT” folder

```
$ cd OUT
$ ls
stdout.txt  H2O.DRHO  H2O.RHO  ...
```

- stdout.txt*:

SCF iteration log

```
605 | | iscf  Eharris(eV)  E_KS(eV)  FreeEng(eV)  dDmax  Ef(eV)  dHmax(eV)
606 | scf: 1 -456.456468 -460.715387 -460.715387 1.588972 -4.556410 12.196431
607 | timer: Routine, Calls, Time, % = IterSCF 1 0.396 27.20
608 | scf: 2 -461.591414 -461.160665 -461.160665 0.026119 -4.119021 10.903015
609 | scf: 3 -463.212274 -462.854178 -462.854178 0.208920 -0.382196 0.972462
610 | scf: 4 -460.450032 -462.025498 -462.025498 0.151698 -2.727828 7.468204
611 | scf: 5 -463.239911 -462.893890 -462.893890 0.124122 -0.244908 0.081254
612 | scf: 6 -462.892597 -462.893549 -462.893549 0.004148 -0.324913 0.171588
613 | scf: 7 -462.894186 -462.894010 -462.894010 0.002377 -0.264588 0.006097
614 | scf: 8 -462.893987 -462.894001 -462.894001 0.000238 -0.260532 0.023874
615 | mix: Pulay -- inversion failed, > SVD [rank/size] 4 / 7
616 | scf: 9 -462.893971 -462.893999 -462.893999 0.000766 -0.276138 0.027002
617 | mix: Pulay -- inversion failed, > SVD [rank/size] 4 / 8
618 | scf: 10 -462.894001 -462.894000 -462.894000 0.000016 -0.275713 0.025725
619 | mix: Pulay -- inversion failed, > SVD [rank/size] 4 / 9
620 | scf: 11 -462.894002 -462.894001 -462.894001 0.000022 -0.275189 0.024089
621 | mix: Pulay -- inversion failed, > SVD [rank/size] 4 / 9
622 | scf: 12 -462.894011 -462.894006 -462.894006 0.000126 -0.272711 0.015711
623 | mix: Pulay -- inversion failed, > SVD [rank/size] 3 / 9
624 | scf: 13 -462.894011 -462.894010 -462.894010 0.000255 -0.267637 0.001180
625 | mix: Pulay -- inversion failed, > SVD [rank/size] 3 / 9
626 | scf: 14 -462.894010 -462.894010 -462.894010 0.000020 -0.267803 0.000885
```

Summary of (energetic) results

```
1693 | siesta: Final energy (eV):
1694 | siesta: Band Struct. = -105.960214
1695 | siesta: Kinetic = 332.398047
1696 | siesta: Hartree = 492.664953
1697 | siesta: Edftu = 0.000000
1698 | siesta: Eso = 0.000000
1699 | siesta: Ext. field = 0.000000
1700 | siesta: Exch. = -99.311166
1701 | siesta: Corr. = -12.541166
1702 | siesta: Bulk bias = 0.000000
1703 | siesta: Exch.-corr. = -111.852332
1704 | siesta: Ion-electron = -1277.145428
1705 | siesta: Ion-ion = 99.146218
1706 | siesta: Ekinion = 0.000000
1707 | siesta: D3 dispersion = 0.000000
1708 | siesta: Total = -464.788542
1709 | siesta: Fermi = -5.446292
```

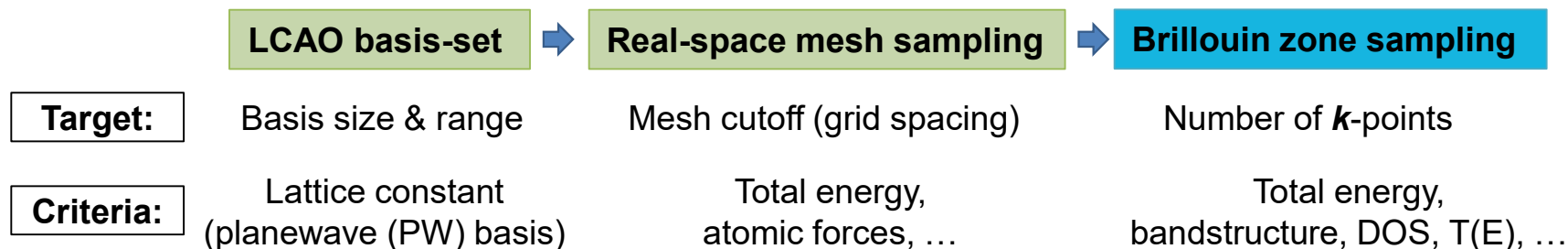
총 에너지, 페르미 준위

Convergence test

➤ [Wiki Website](#)

k -point 샘플링을 통한 총-에너지 수렴성 테스트 과정을 알아본다

Convergence tests: 신뢰도 있는 전자구조 계산을 위한 “수치적” 수렴성 테스트 과정



- Change the directory to the “2.convergence_test/” folder:

```
$ cd ../../../../2.convergence_test/
$ ls
```

basis_set/ grid_sampling/

kpoints_sampling/

Homeworks

Today's exercise

* 일반적으로:

Basis set size는 DZP 수준으로 충분

Basis range (energy shift)와

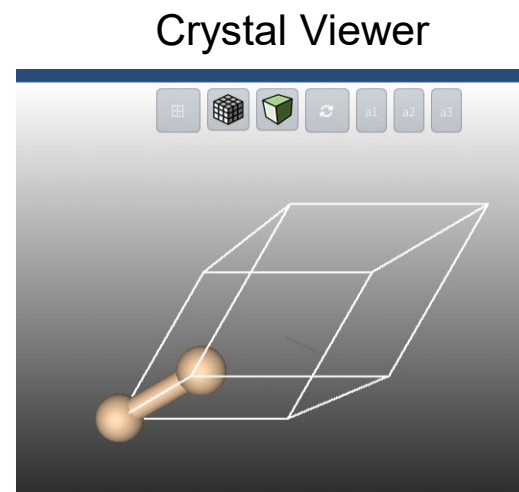
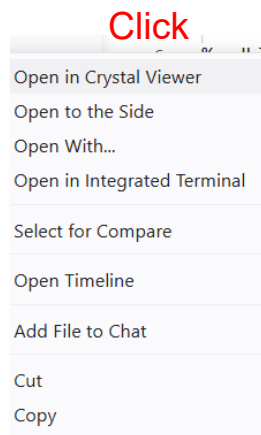
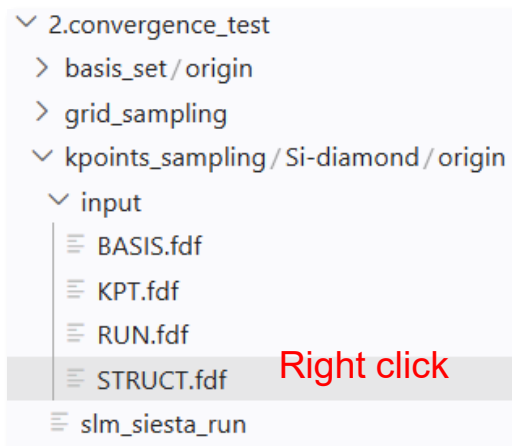
Grid spacing (mesh cutoff)

계산 시스템 / 수준 / 종류에 따라서 적절한 선택 필요

- Change the directory to the “kpoints_sampling/Si-diamond” folder:

```
$ cd kpoints_sampling/Si-diamond  
$ ls  
origin/
```

- Check the atomic geometry (origin/input/STRUCT.fdf):



- Generate the calculation cases with different *k*-points:

```
$ py4siesta
Select menu number: 1

[Bulk K-point sampling]
Press Enter on an empty line to finish.
Input k-point value #1: 1
Input k-point value #2: 3
Input k-point value #3: 5
Input k-point value #4: 7
Input k-point value #5: 10
Input k-point value #6: 15
Input k-point value #7: 20
Input k-point value #8:
```

- Check the generated calculation directories:

```
$ ls 01.kpoint_sampling/
001+001+001/  003+003+003/  005+005+005/  007+007+007/
010+010+010/  015+015+015/  020+020+020/
```


- Check the difference:

```
$ diff 01.kpoint_sampling/001+001+001/input/KPT.fdf
01.kpoint_sampling/020+020+020/input/KPT.fdf

2,4c2,4
<      1      0      0      0.000000
<      0      1      0      0.000000
<      0      0      1      0.000000
---
>     20      0      0      0.000000
>      0     20      0      0.000000
>      0      0     20      0.000000
```

- Job submission:

```
$ py4siesta
Select menu number: 9
```

- After completing all calculations, printout the total energy for each case:

```
$ grep "Total =" 01.kpoint_sampling/*/OUT/stdout.txt

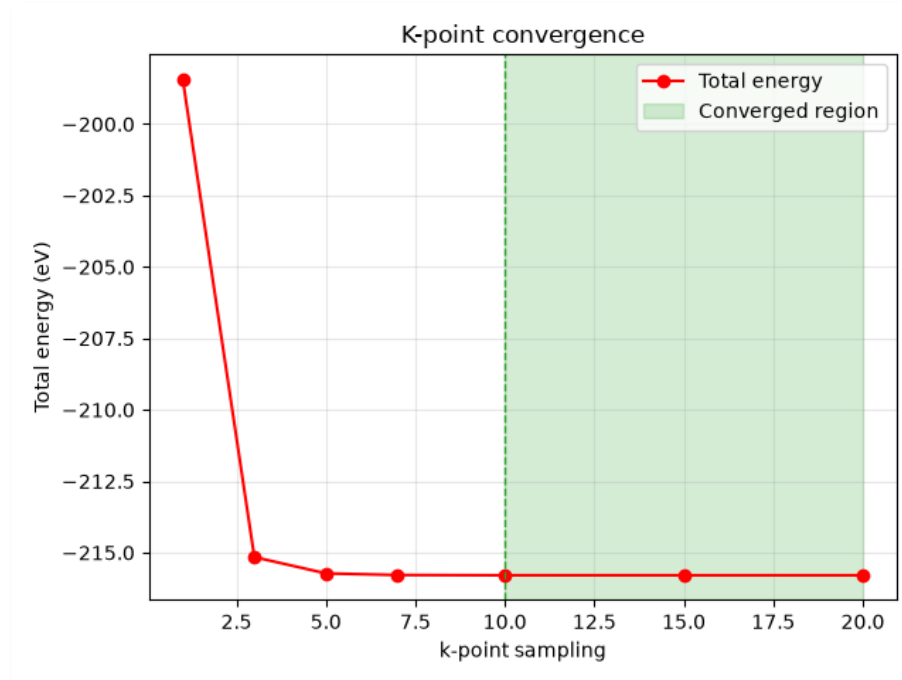
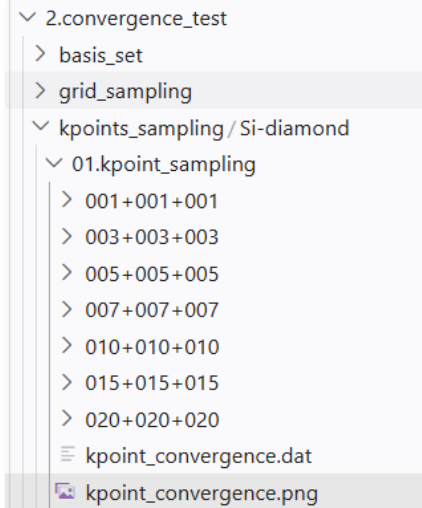
grep 'Total =' 01.kpoint_sampling/*/stdout.txt
01.kpoint_sampling/001+001+001/stdout.txt:siesta:          Total =      -198.430842
01.kpoint_sampling/003+003+003/stdout.txt:siesta:          Total =      -215.155253
01.kpoint_sampling/005+005+005/stdout.txt:siesta:          Total =      -215.726530
01.kpoint_sampling/007+007+007/stdout.txt:siesta:          Total =      -215.783615
01.kpoint_sampling/010+010+010/stdout.txt:siesta:          Total =      -215.792939
01.kpoint_sampling/015+015+015/stdout.txt:siesta:          Total =      -215.793765
01.kpoint_sampling/020+020+020/stdout.txt:siesta:          Total =      -215.793788
```

- Total energy convergence analysis from python script:

```
$ py4siesta
Select menu number: 3

[K-point convergence analysis]
Input total energy tolerance in eV [default: 0.01]:
Converged at k = 10 with tolerance 0.010000 eV
```

- Results:



10×10×10 *k*-points sampling 이후에 총 에너지가
0.01 eV 이하로 수렴 → 이후 구조 최적화 및
전자구조 계산 시에 해당 수렴된 *k*-points 사용

Geometrical optimization

➤ [Wiki Website](#)

구조 최적화 과정을 알아본다

Geometrical optimization: 안정한 평형상태(원자힘 최소화)의 원자구조를 계산하는 과정

Lattice vector optimization



Basis atoms relaxations

- Change the directory to the “3.geometrical_optimization/” folder:

```
$ cd ../../../../3.geometrical_optimization/Si-diamond
```

- Check the initial structure files (origin/input/STRUCT.fdf):

```
#(3) Lattice, coordinates, k-sampling

LatticeConstant      1.000000000 Ang
%block LatticeVectors
  3.827985222      0.000000000      0.000000000
  1.913992611      3.315132448      0.000000000
  1.913992611      1.105044149      3.125536846
%endblock LatticeVectors

AtomicCoordinatesFormat Ang
%block AtomicCoordinatesAndAtomicSpecies
  0.000000000      0.000000000      0.000000000      1      1
  1.913992611      1.105044149      0.781384212      1      2
%endblock AtomicCoordinatesAndAtomicSpecies
```

단위격자 벡터

Basis 원자 위치

- Generate the calculation cases for different lattice constants:

```
$ py4siesta
Select menu number: 4
[Bulk structure optimization]
Input bulk expansion direction (x y z) [default: 1 1 1]:
```

- Check the difference:

```
$ diff 02.volume_eos/06-1.000/input/STRUCT.fdf
02.volume_eos/11-1.010/input/STRUCT.fdf

12,14c12,14
<      3.827985222      0.000000000      0.000000000
<      1.913992611      3.315132448      0.000000000
<      1.913992611      1.105044149      3.125536846
---
>      3.866265074      0.000000000      0.000000000
>      1.933132537      3.348283772      0.000000000
>      1.933132537      1.116094590      3.156792214
20c20
<      1.913992611      1.105044149      0.781384212      1      2
---
>      1.933132537      1.116094590      0.789198054      1      2
```

단위격자 벡터
 $3.156792214/3.125536846 = 1.10$

Basis 원자 위치

- Job submission:

```
$ py4siesta  
Select menu number: 10
```

- After completing all calculations, printout the total energy for each case:

```
$ grep 'Total =' 02.volume_eos/*/stdout.txt  
  
02.volume_eos/01-0.990/stdout.txt:siesta:          Total =      -215.791013  
02.volume_eos/02-0.992/stdout.txt:siesta:          Total =      -215.794351  
02.volume_eos/03-0.994/stdout.txt:siesta:          Total =      -215.796772  
02.volume_eos/04-0.996/stdout.txt:siesta:          Total =      -215.798292  
02.volume_eos/05-0.998/stdout.txt:siesta:          Total =      -215.798930  
02.volume_eos/06-1.000/stdout.txt:siesta:          Total =      -215.798703  
02.volume_eos/07-1.002/stdout.txt:siesta:          Total =      -215.797627  
02.volume_eos/08-1.004/stdout.txt:siesta:          Total =      -215.795718  
02.volume_eos/09-1.006/stdout.txt:siesta:          Total =      -215.792996  
02.volume_eos/10-1.008/stdout.txt:siesta:          Total =      -215.789481  
02.volume_eos/11-1.010/stdout.txt:siesta:          Total =      -215.785185
```

- EOS for bulk materials → **Murnaghan equation of state**¹

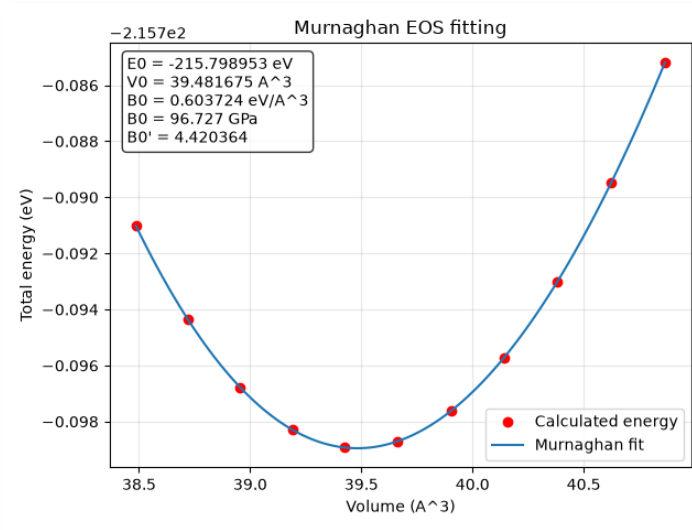
$$E(V) = E_0 + \frac{B_0 V}{B_0'} \left[\frac{(V_0/V)^{B_0'}}{B_0' - 1} + 1 \right] + \frac{B_0 V}{B_0'}$$

Energy	$E = E^{total}$
Pressure	$P = -\frac{dE}{dV}$
Bulk Modulus	$B = -V \frac{dP}{dV} = V \frac{d^2E}{dV^2}$

- Perform the fitting analysis:

```
$ py4siesta
Select menu number: 8
Select fitting mode: 1
```

```
3.geometrical_optimization / Si-diamond
  02.volume_eos
    > 01-0.990
    > 02-0.992
    > 03-0.994
    > 04-0.996
    > 05-0.998
    > 06-1.000
    > 07-1.002
    > 08-1.004
    > 09-1.006
    > 10-1.008
    > 11-1.010
    > optimized_structure
    {} eos_config.json
    ≡ eos_fitting_parameters.dat
    eos_fitting.png
```



- Change the directory to the generated optimal structure (from previous EOS):

```
$ cd 02.volume_eos/optimized_structure
```

]

- Turn on the atomic relaxation option :

```
$ vi input/RUN.fdf
```

```
MD.TypeOfRun      CG          # Type of dynamics:
MD.VariableCell    F
MD.NumCGsteps      100         # 0 for single-point / 100 for gopt
MD.MaxForceTol     0.02 eV/Ang # Default: 0.04 eV/Ang
```

- Slightly modify the atomic coordinate (first basis Si atom):

```
$ vi input/STRUCT.fdf
```

```
(origin)  0.000000000  0.000000000  0.000000000  1  1  ==>
(revised) 0.000000000  0.000000000  0.100000000  1  1
```

- Job submission:

```
$ sbatch slm_siesta_run
```

- Check the “atomic relaxation process” in stdout.txt:

```
$ grep "CG opt." OUT/stdout.txt
Begin CG opt. move =      0
...
Begin CG opt. move =      3
Begin CG opt. move =      4

$ grep "constrained" OUT/stdout.txt
Max      1.171653      constrained
...
Max      1.306183      constrained
Max      0.014018      constrained << 0.02 eV/ang
```

- Obtain final optimized atomic geometry

```
$ cd OUT
$ ls
Si.STRUCT_OUT
(conda542) $ conda deactivate
(base)$ structout2fdf
Si.STRUCT_OUT → STRUCT.fdf
(base)$ diff STRUCT.fdf ../input/STRUCT.fdf
(base)$ conda activate conda542
(conda542) $ conda deactivate
```

Electronic structure analysis

➤ [Wiki Website](#)

전자구조 계산 및 분석 과정을 알아본다

Electronic structure analysis: 수렴된 계산 조건 및

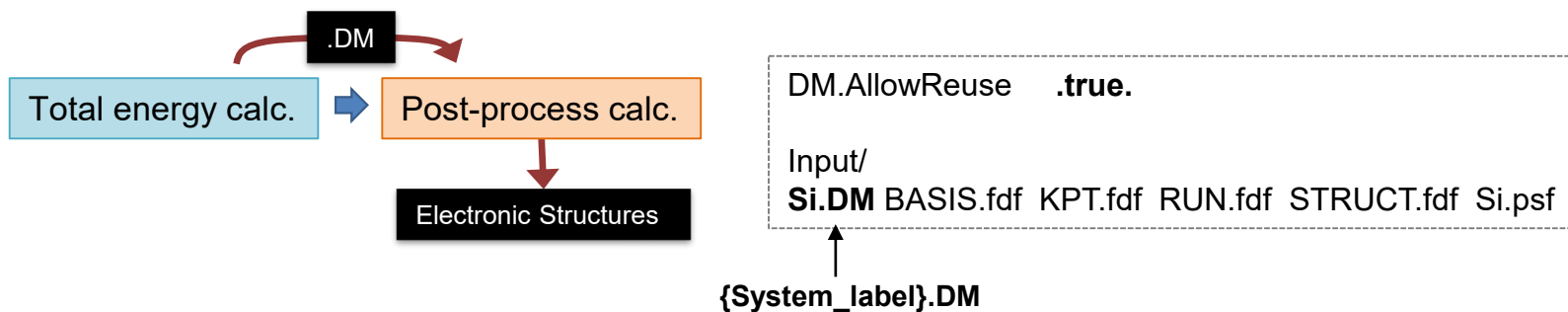
Band structure

(Projected) density of states

Bond analysis

- Change the directory to the “4.electronic_structures/” folder:

```
$ cd ../../../../4.electronic_structures/
$ ls
1.band_structure/  2.density_of_states/
3.bond_analysis/
```

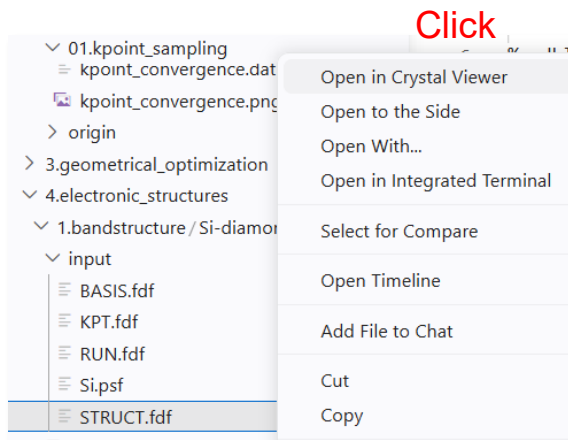


- Change the directory to the “1.band_structure/Si-diamond”:

```
$ cd 1.band_structure/Si-diamond/
```

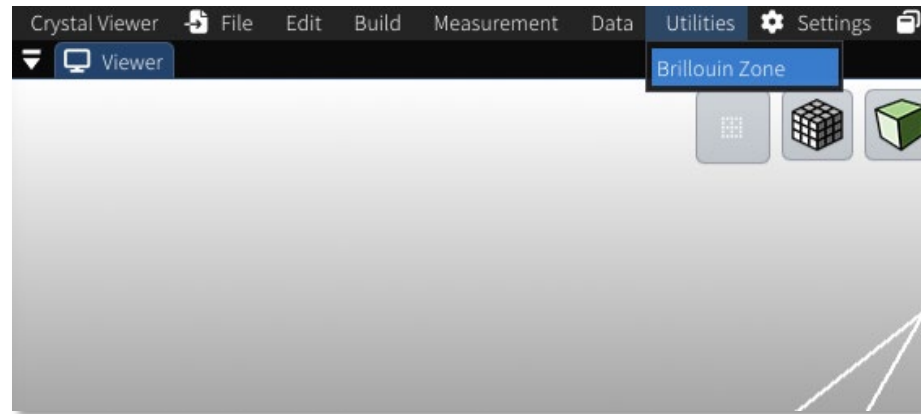
- Show Brillouin zone with the Crystal Viewer:

(1) Open Crystal Viewer



Right click

(2) Select [Utilities] / [Brillouin Zone]



Exercise: (1) Band structure - Silicon

- Check the band k -path along the high symmetry k-points (L-G-X-U)

Bandpath Configuration:

Path:

Npoints: - +

Options:

☒ Show reciprocal vectors

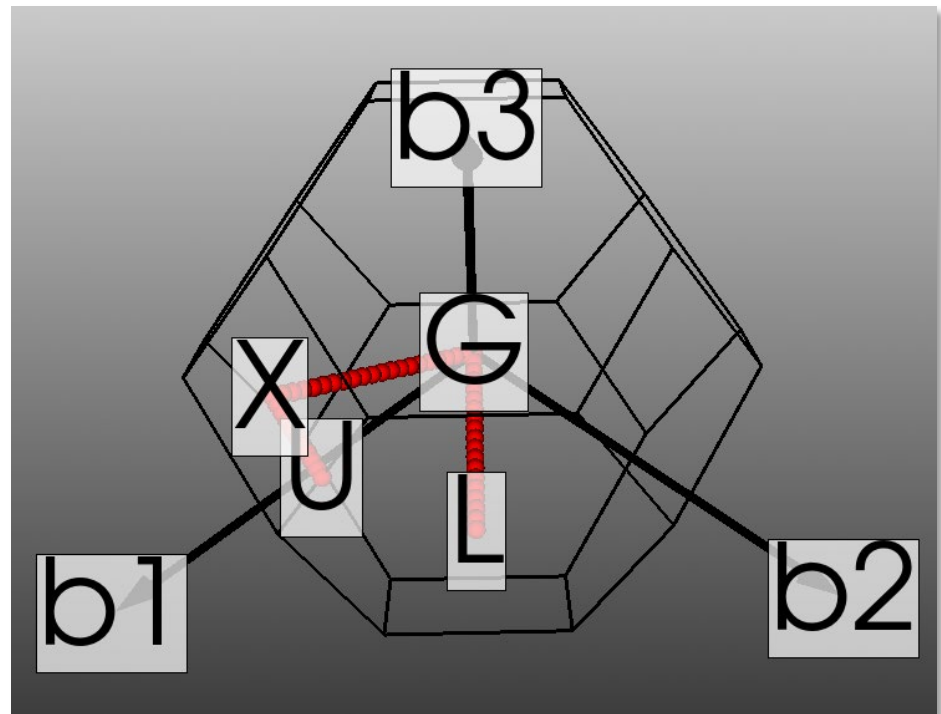
☒ Show special point labels

? BZ Plot Mode (vectors: ON, labels: ON)

Special k-points for lattice: fcc

Total points: 6

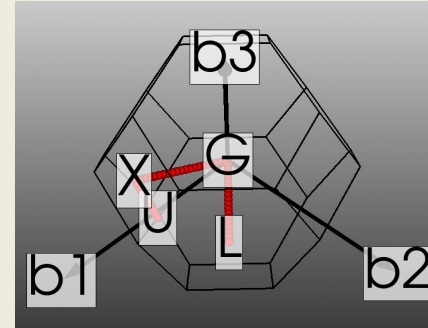
Label	a	b	c
G	0.000000	0.000000	0.000000
K	0.375000	0.375000	0.750000
L	0.500000	0.500000	0.500000
U	0.625000	0.250000	0.625000
W	0.500000	0.250000	0.750000
X	0.500000	0.000000	0.500000



- Check the band k -path option and submit the job:

```
$ vi input/RUN.fdf
...
BandLinesScale ReciprocalLatticeVectors
%block BandLines
  1    0.50000 0.500000 0.50000  L
  50   0.00000 0.000000 0.00000  G
  50   0.50000 0.000000 0.50000  X
  50   0.62500 0.250000 0.62500  U
%endblock BandLines

$ sbatch slm_siesta_run
```

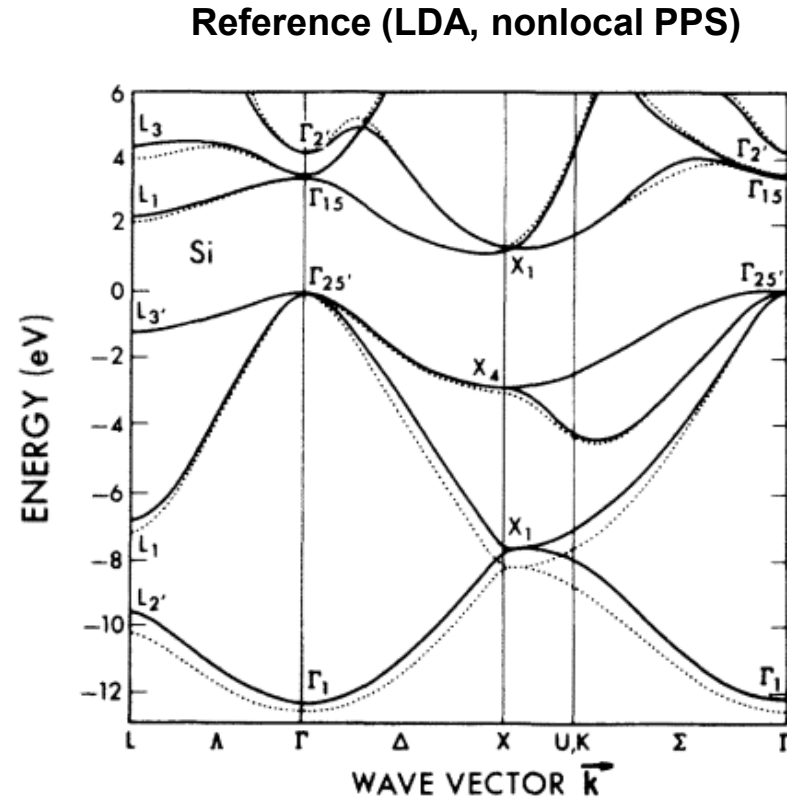
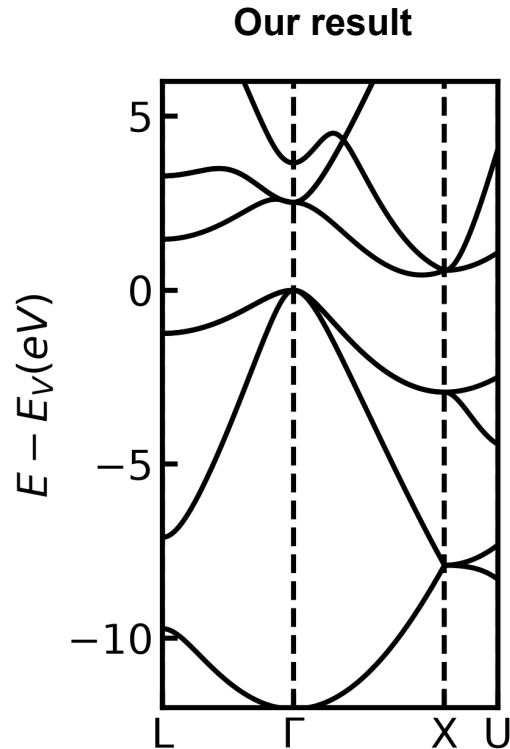


- Plot the band structure:

```
$ cd OUT/
$ py4siesta-tool band --emin -12 --emax 6
```

Exercise: (1) Band structure - Silicon

- Results:



Phys. Rev. B **14**, 556 (1976)

- Change the directory to the “2.density_of_states/GaAs-zincblende”:

```
$ cd ../../2.density_of_states/GaAs-zincblende
$ vi input/RUN.fdf

%block PDOS.kgrid_Monkhorst_Pack
  30  0  0  0
  0  30  0  0
  0  0  30  0
%endblock PDOS.kgrid_Monkhorst_Pack

%block ProjectedDensityOfStates
  -20.0  10.00  0.15  300 eV
%endblock ProjectedDensityOfStates
```

k-points sampling for PDOS analysis
(보통 전자구조 계산 k-points보다 더 많은 값을 줌)

{Emin} {Emax} {Smearing} {number of points} {unit}

Density of states (DOS): $g(E) = \sum_{i,k} \delta(E - E_{i,k})$

Projected DOS: $g_{\mu}(E) = \sum_{i,k} \sum_{\nu} c_{i\mu} c_{i\nu} S_{\mu\nu} \delta(E - E_{i,k})$

$$\delta(E - E_{i,k}) \approx \frac{1}{\sigma\sqrt{\pi}} e^{-\frac{(E-E_{i,k})^2}{\sigma^2}}$$

- Job submission:

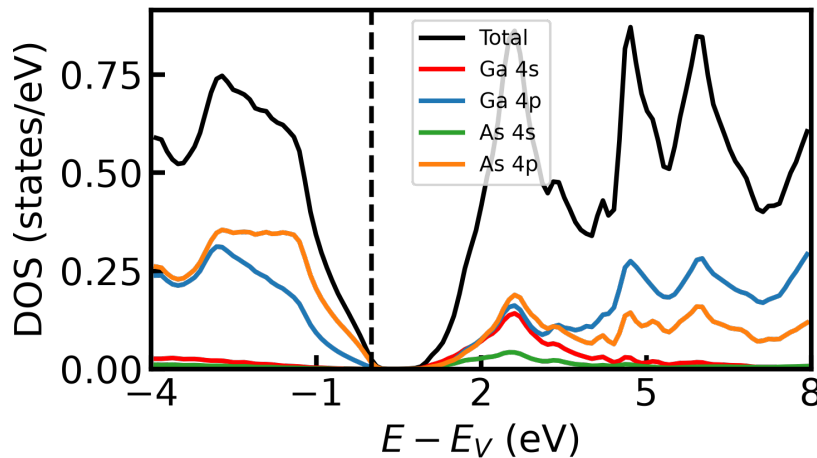
```
$ sbatch slm_siesta_run
```

- Plot the density of states:

```
$ cd OUT  
$ py4siesta-tool pdos --emin -4 --emax 8 --orbital  
Ga_4_0 Ga_4_1 As_4_0 As_4_1
```

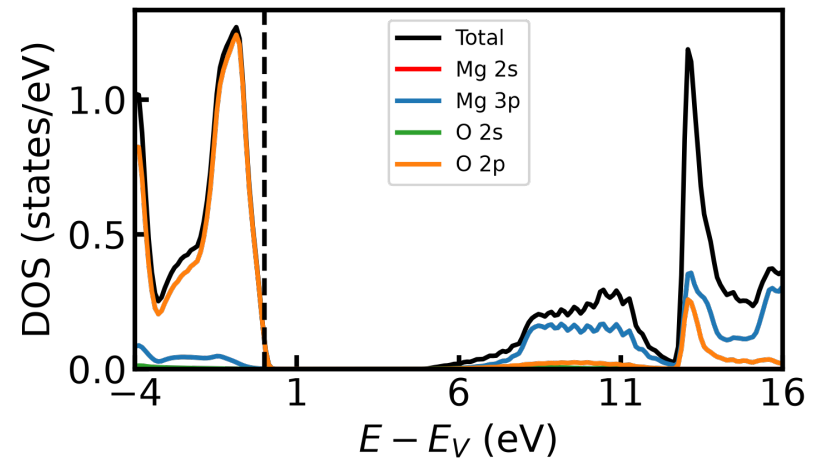
- Results:

GaAs PDOS



MgO PDOS

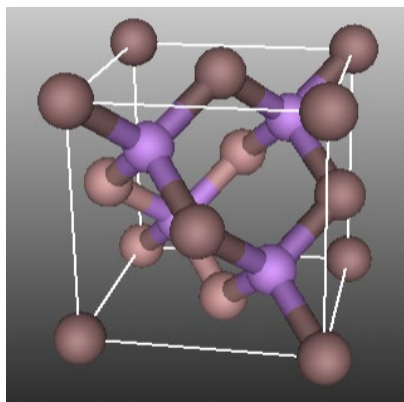
(Homework, 2.density_of_states/MgO-rocksalt)



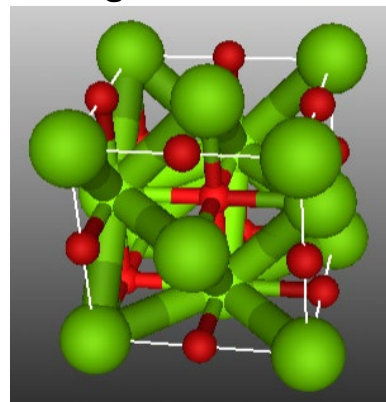
- Change the directory to the “3.bond_analysis

```
$ cd ../../3.bond_analysis  
$ ls  
GaAs-zincblende  MgO-rocksalt
```

GaAs-zincblende



MgO-rocksalt



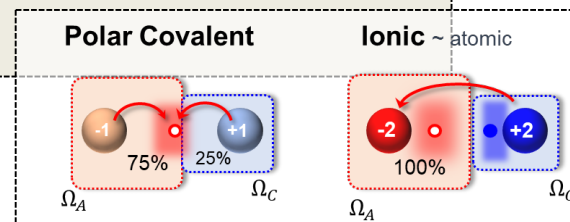
- Check the options and perform the calculations** for each case:

```
$ vi GaAs-zincblend/input/RUN.fdf  
SaveRho                      T  
SaveDeltaRho                  T      # SystemLabel.DRHO  
WriteMullikenPop               1
```

- Check the options and perform the calculations for each case

```
$ vi GaAs-zincblend/input/RUN.fdf

#(10) Options for saving/reading information
SaveRho                T
SaveDeltaRho           T      # SystemLabel.DRHO
WriteMullikenPop        1
```



- Mulliken population analysis:

GaAs-zincblend/OUT/stdtout.txt

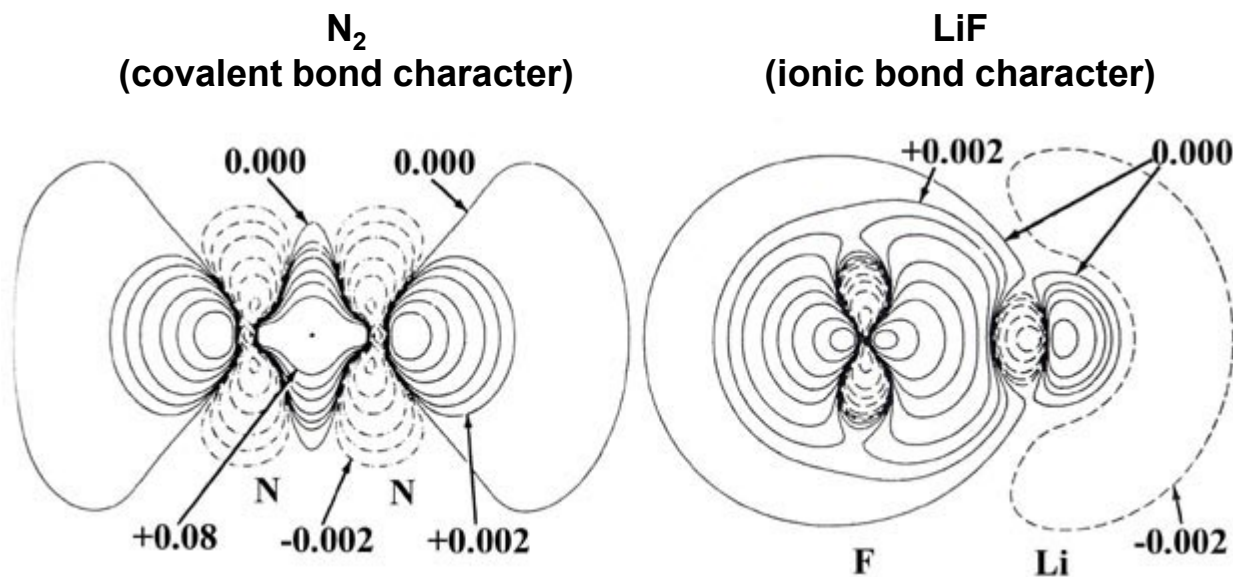
Atom #	charge [q]	valence [e]	Species
1	-0.536940	13.536940	Ga
2	-0.536940	13.536940	Ga
3	-0.536940	13.536940	Ga
4	-0.536940	13.536940	Ga
5	0.536940	14.463060	As
6	0.536940	14.463060	As
7	0.536940	14.463060	As
8	0.536940	14.463060	As
Total	-0.000000		

MgO-rocksalt/OUT/stdtout.txt

Atom #	charge [q]	valence [e]	Species
1	1.177354	0.822646	Mg
2	1.177354	0.822646	Mg
3	1.177354	0.822646	Mg
4	1.177354	0.822646	Mg
5	-1.177354	7.177354	O
6	-1.177354	7.177354	O
7	-1.177354	7.177354	O
8	-1.177354	7.177354	O
Total	0.000000		

Cf. More accurate approach: **Bader charge analysis** from all electron density distributions

Dr. Richard F.W. Bader, <An Introduction to the Electronic Structure of Atoms and Molecules>



Deformation electron density:
$$\delta\rho(\mathbf{r}) = \rho(\mathbf{r}) - \sum \rho^{atom}(\mathbf{r})$$

Total electron density Neutral atomic density

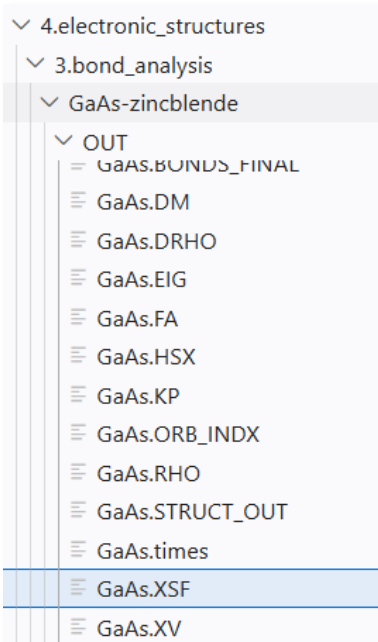
- Change the directory to the “OUT” for GaAs case

```
$ cd GaAs-zincblend/OUT
$ rho2xsf < ../rho2xsf_GaAs.txt
$ ls
... GaAs.XSF
```

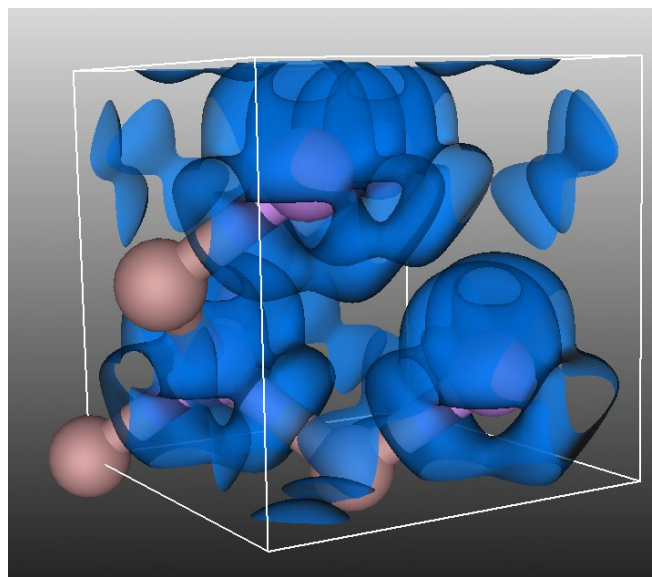
직접 CLI 입력도 가능

```
GaAs
A
0 0 0
5.629831240 0.000000000 0.000000000
0.000000000 5.629831240 0.000000000
0.000000000 0.000000000 5.629831240
96 96 96
DRHO
RHO
BYE
```

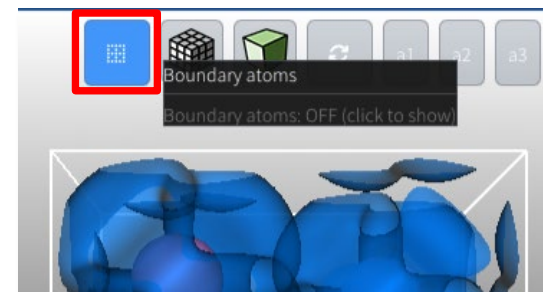
Vscode explorer



Crystal Viewer



Select “Boundary atoms”

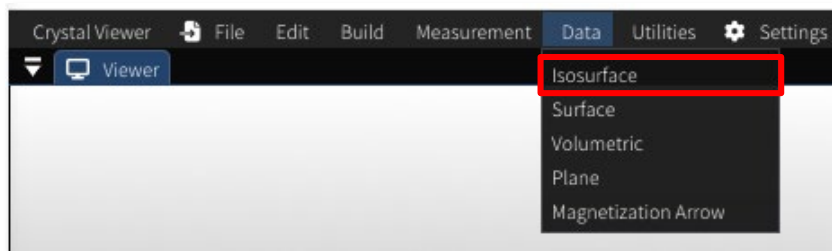


$$\delta\rho(\mathbf{r}) = \rho(\mathbf{r}) - \sum \rho^{atom}(\mathbf{r})$$

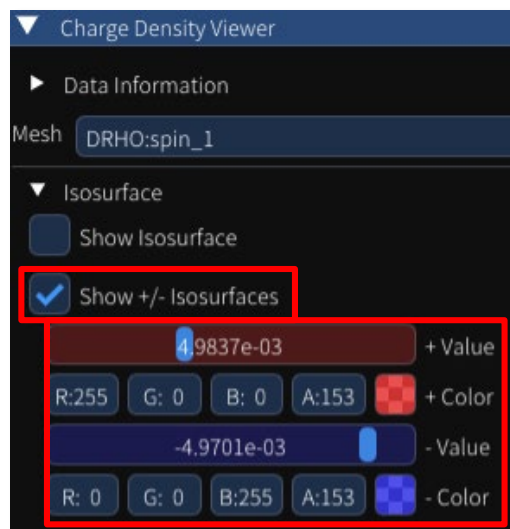
→ Charge redistribution after chemical bonding

- Show isosurface:

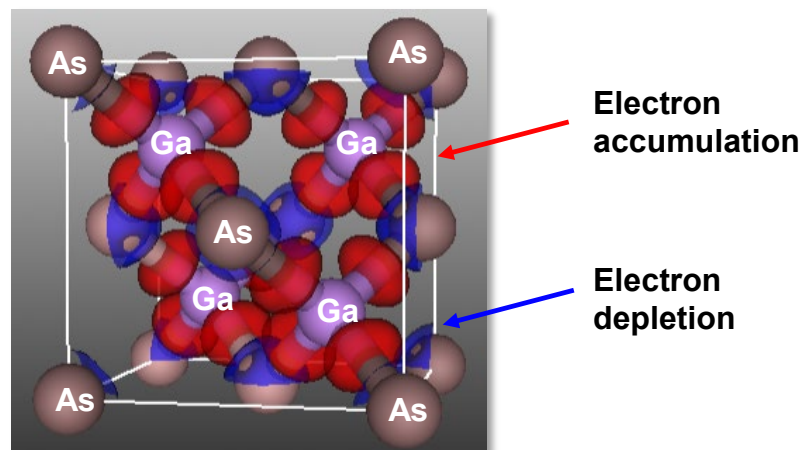
(1) Click **[Data] / [Isosurface]**



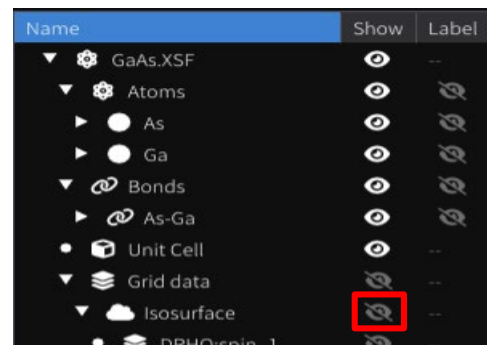
(2) Disable **[Show Isosurface]** and enable **[Show +/- Isosurfaces]**



(3) Analyze the results:

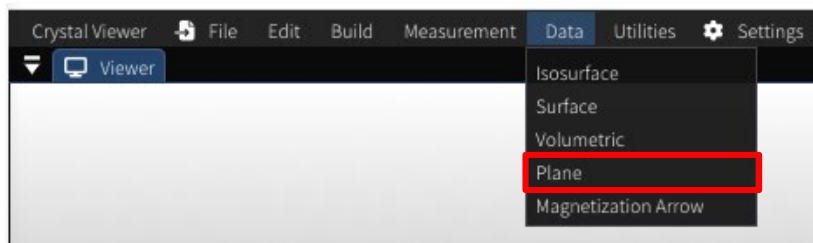


(+) Disable the **[Isosurface]** for next analysis

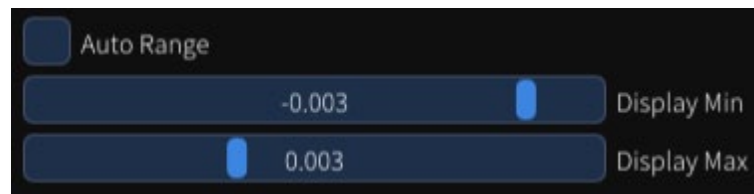


- Show slice plane:

(1) Click **[Data] / [Plane]**



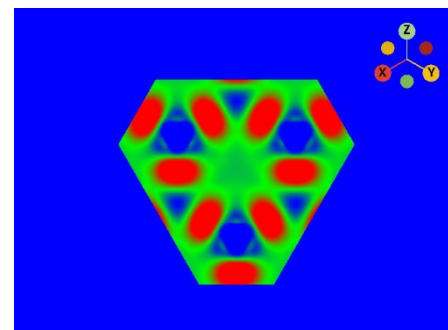
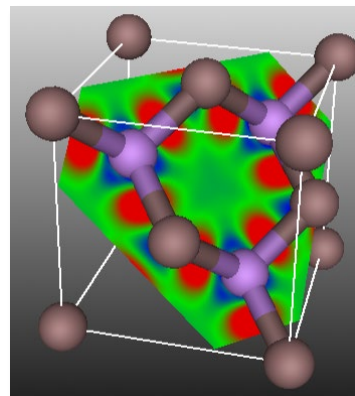
(2) In 2D Slice viewer, disable **[Auto Range]** and set the color range by **$[-3e^{-3}, 3e^{-3}]$**



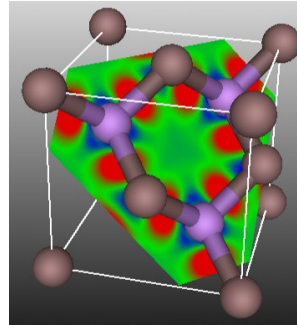
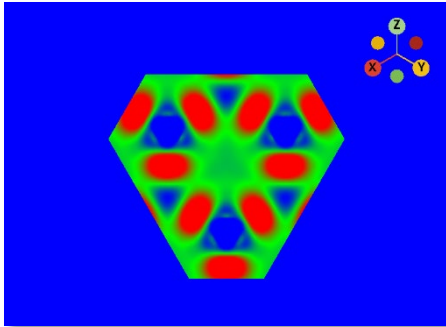
(3) Select the **[Define plane by Miller indices]**, and set the **(h,k,l)** by **(1,1,1)**, then modify the **[Position]**



(4) Analyze the results:

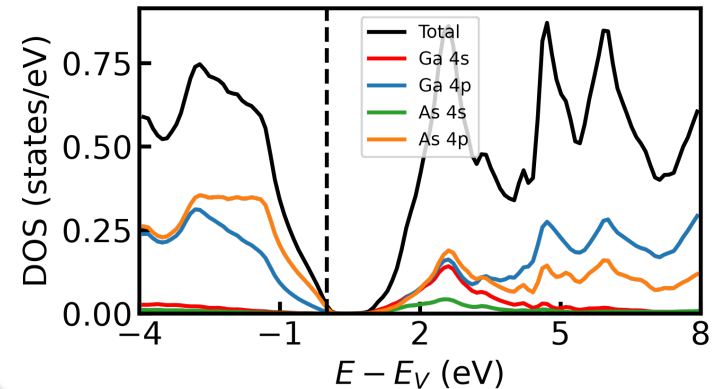


GaAs

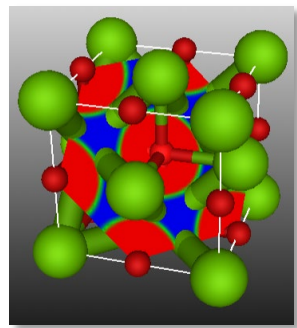
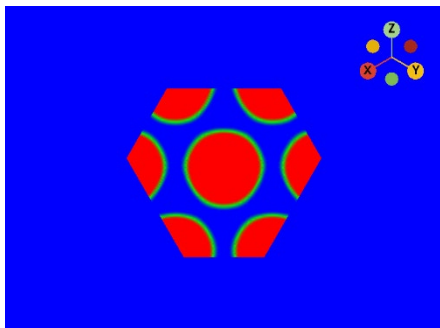


$$\delta\rho(\mathbf{r}) = \rho(\mathbf{r}) - \sum \rho^{atom}(\mathbf{r})$$

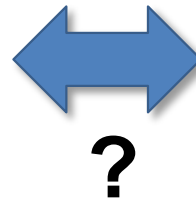
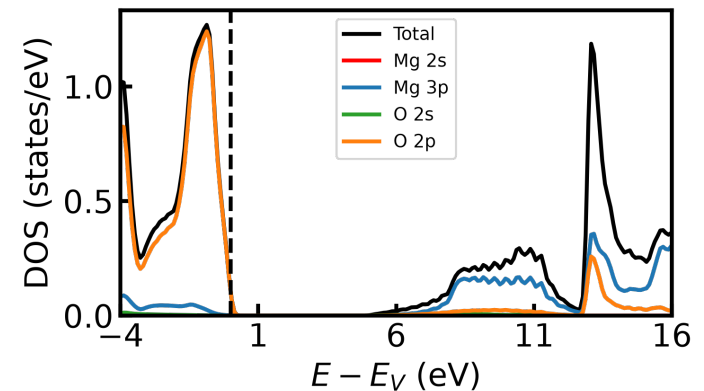
PDOS



MgO



PDOS



화학 결합 특성과 전자구조 간의 상관관계?



- **Modeling of (initial) atomic structure:**
전자구조 특성 계산의 대상인 원자 구조(원소, 위치)를 모델링하는 과정
- **Basis-set optimization:**
DFT 계산의 LCAO 기저함수의 최적화 과정;
- **Brillouin zone (BZ) (or k -points) sampling :**
주기적 경계조건에 따른 Brillouin zone (BZ) 에 대한 수치적인 k -points 샘플링 과정
- **Geometrical optimization:**
총 에너지, 원자힘 계산을 통한 안정한 구조를 찾는 과정; 격자 상수(lattice constant) 및 원자 위치
- **Electronic structure analysis:**
최종적으로 얻은 계산 조건 및 구조를 통해 밴드구조, 상태밀도함수 등 전자구조 계산 및 분석 과정

Advanced topic: [Projected band structure](#), [Phonon dispersion](#), [Quantum transport](#), [Optical properties](#), Etc